

Key Takeaways: Roles and Tools

Estimated time needed: 5 minutes

Introduction

The Role and Tools video focused on four aspects of a cybersecurity architect. These include:

- Role
- Mindset
- Tools
- Domains

Here are the key takeaways.

Role and Mindset

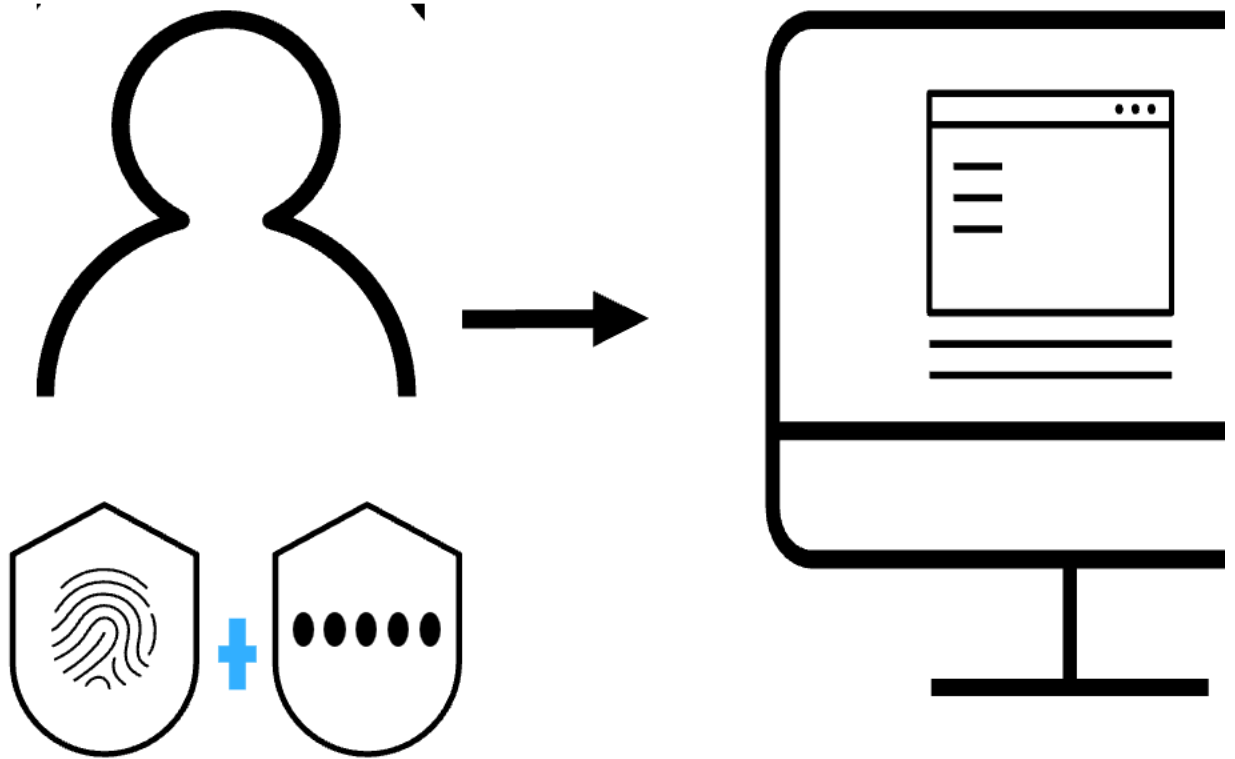
Alt all starts with stakeholders providing their requirements to the architect. The architect will develop reference architecture diagrams that show the relations between the high-level components. The architecture diagrams will be translated into an IT architecture diagram.

A simple architecture diagram with a user accessing a device (Desktop/laptop/mobile), which then connects to a web server, which in turn connects with the Application server and accesses a database to retrieve information.

The architect then transfers the information to the engineer, who will implement the system.

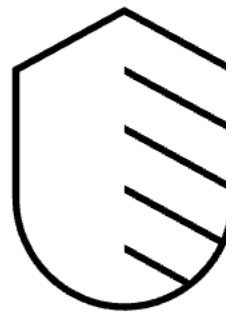
Tip: Architect = Whiteboard; Engineer = Keyboard

An architect must understand how a system works to assess how and where the system might fail. Always ask: **What do I do to prevent the system from failing?** And then figure out how to **prevent the failure**.



**Multi-Factor
Authentication
(MFA)**

**Mobile Device
Management
(MDM)**



**Endpoint
Detection**



Note: Different types of engineers may implement different parts of the architecture. For example, a Database administrator will implement the database, while the network engineer will implement firewalls.

Commonly Used Tools of the Trade for an Architect

In the case of IT, an architect is creating diagrams. The diagrams act as tools for the architects, and the most common ones are:

- **Business Context diagrams** provide a high-level line of business view. They show the relationship between different entities in a system.
- **System Context diagrams** decompose the business context diagram further to determine how the entities might look in a system.
- **Architecture Overview diagrams** decompose the system context diagrams to determine how the smallest components will look and connect with other components.

The diagrams are part of the architect's vocabulary. They should be designed so that any IT architect could understand them.

A cybersecurity architect will study the diagrams created by the IT architect and determine how to make the systems and overall architecture secure and fail-safe.

The cybersecurity architect will need to use the following as a checklist:

- Five principles of cybersecurity
- CIA triad
- Frameworks, such as NIST CSF, provides a checklist to ensure that you have covered all your bases

As a cybersecurity architect, it is your job to apply all the items in the checklist to the different diagrams and add security components.

It is important to involve the cybersecurity architects in the Risk Analysis phase of a project rather than in the implementation phase. For a secure design and implementation, applying the cybersecurity principles upfront and creating a security policy is important. Then, follow it with an architecture that integrates well with the overall IT architecture.

Domains

Each component in the architecture is a domain. For example, in the graphic, each component, like the user, desktop, and network, are all domains that need to be secured from attackers.

In addition, a monitoring system that monitors all the domains is also a domain.

If there is a problem, a response is needed to ensure that the problem is resolved quickly.

Author

Dee Dee Collette



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